Development effects of electrification in Brazil

Identification method and replication of main results of *Lipscomb et al (2013)*

Paula Luvini, Rodrigo Lugo

2026-10-29

Research design: ideal benchmark vs. the paper

- **Ideal design**: Among areas where grid expansion is feasible,a lottery decides which get the grid now and which wait ten years.
- **The paper**: Builds a county and decade FE-IV (2SLS). The instrument (for actual electricity availability) is a *simulated grid* built from geographic characteristics and each decade's budget.

Causal interpretation only if

Relevance of the instrument: met, first-stage $F \approx 25 > 10$.
Exogeneity of the instrument: no relationship between predicted electricity and confounders (Table 10).
Exclusion restriction: geography affects local development only via electrification.
Monotonicity: not discussed but very plausible to hold.

Main findings

Preferred IV specification (county and decade FE, Amazon $\times$ decade dummies):

- **Housing values**: +10 pp electrification $\rightarrow$ **+881 reais**. It's a change of +6.8% of the mean value (Table 6).
- **HDI**: moving from 0 to full electrification $\rightarrow$ **+0.11**. Would move the median county towards the 69th percentile county (Table 7).

Mechanisms: electrification appears to improve development through education and employment gains, with limited evidence that migration explains the results and no effect on health.

Replication of main results

Is the design convincing?

Convincing

The instrument simulates electricity expansion using hydropower geography and decade-specific budgets, leaving out local demand. Results remain robust after additional controls for the geographic factors used to construct the instrument (Table 10).

Not convincing

The instrument predicts dams: they bring construction, roads, flood control and ecosystem change. This could bring positive (or negative!) effects on economic development, unrelated to electrification. Could this risk the exclusion restriction?

Appendix: Table 6

Appendix: Table 7

Appendix: Table 10